

# Devin Abraham

Chicago, IL • 312-669-0017 • [dja6@rice.edu](mailto:dja6@rice.edu) • <https://tinyurl.com/devin-portfolio>

## EDUCATION

### Rice University

B.S. Operations Research, Minor Data Science  
GPA: 3.85

Houston, TX

May 2027

## EXPERIENCE

### Data Science/Engineering Intern

May 2026 - August 2026

CCC Intelligent Solutions, Chicago, IL

- Developed a capacitated facility location optimization model to identify high-priority repair shop certification candidates for OEMs nationwide, projecting \$2M in incremental revenue
- Built a disaster-response simulation model using GLMs and Monte Carlo methods to help insurance carriers allocate employees 30% more effectively during catastrophe events

### AI / Backend Software Engineering Intern

May 2025 - August 2025

1905 New Media, Chicago, IL

- Reduced testing time by 95% by developing a Quality Control AI Agent with n8n that auto-generated and executed test cases for each feature of a proprietary AI system
- Redesigned PostgreSQL schema to support user-role and family relations, scheduling, enrollment, attendance tracking, and audit logging, resolving all major client pain points and improving system reliability

### External Vice President

April 2025 - Present

Rice INFORMS, Chicago, IL

- Plan 3 industry events and 2 Academia events for 150 members along with organizing two social events
- Manage communication for 200 INFORMS members, ensuring timely updates and engagement across activities

### Live Instructor

May 2024 - August 2024

Rice ELITE Tech Camp, Houston, TX

- Collaborated with Rice University faculty and co-workers to develop a new Robotic Engineering II curriculum and fine-tune a 3D modeling and 3D printing course; Converted curriculum to daily lesson plans in Canvas

## PROJECTS

### Rice INFORMS

Team Lead, Rice University

January 2026-May 2026

- Led development of mixed-integer program for HARC to allocate tree planting to maximize heat index reduction under \$2 million budget, capacity, and land usage constraints

Team Lead

January 2025-May 2025

- Reduced scheduling time by 90% and increased preferred time slots by 60% by leading a team of 6 to develop a scheduling optimization program for the Rice EMS group
- Launched a web-based scheduling platform using Python and TypeScript, automating calendar invite distribution and streamlining EMS member coordination

### Class Project

January 2026 – May 2026

CMOR 544, Houston, TX

- Created multi-agent reinforcement learning model for the card game Euchre. Implemented a MCTS for the bidding phase and QMIX architecture for trick play
- Wins at a 65% rate against strong rule based model outperforming reproduced models by over 20%

### Data Science / Operations Research Undergraduate Researcher

January 2025 - April 2026

Center for Operations Research in Cancer and MD Anderson, Houston, TX

- Developed a Partially Observed Markov Decision Process (POMDP) model to optimize recurrent cancer screening plans for head and neck patients; scheduled for clinical testing at MD Anderson in 2027

## ADDITIONAL INFORMATION

**Awards:** Chicago Engineers Foundation Scholar, Cum Laude, Erasmus (teacher-nominated society for passionate learners)

**Leadership:** Senator and Treasurer for Duncan College, H&D Food Rep, HS Baseball Varsity Captain

**Skills:** Python, Java, R, SQL, PyTorch, TensorFlow, Sklearn, Spark, AWS, Airflow, Gurobi, Git, Gymnasium, NumPy